

PUBLIC EXHIBIT 13

EXHIBIT 13

Oura Ring 4 Infringement Claim Chart - U.S. Patent No. 10,642,359

	U.S. Patent No. 10,642,359
Independent Claim 1	Oura Ring 4 and Corresponding Applications (collectively, “Oura Accused Product”)
1[pre] 1. A wearable device comprising:	To the extent that the preamble is limiting, the Oura Accused Product is a wearable device.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

<https://ouraring.com/blog/oura-ring-4/>

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ) and [Oura Ring Generation 3](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ).
(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
1[a] a motion sensor;	The Oura Accused Product includes a motion sensor. For example, the Oura Accused Product includes a motion sensor that measures motion, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4)

U.S. Patent No. 10,642,359			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	
	https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4		
1[b] a biological sensor; and	The Oura Accused Product includes a biological sensor. For example, the Oura Accused Product includes a biological sensor, such as a sensor that measures biological information such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate. These sensors include red, green, and/or IR LEDs and sensors.		

	U.S. Patent No. 10,642,359
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4

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How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

(<https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2>)

What Is Live Heart Rate?

With Live Heart Rate, you can check your heart rate at any time during the day. Just tap to see your heart rate instantly. You can access Live Heart Rate through the 24-hour [heart rate graph](#).

Oura uses green LEDs to measure your heart rate in real-time. The process uses photoplethysmography (PPG) technology at a frequency of 50Hz.

PPG technology works by shining a light onto the surface of your skin. When this reaches your blood vessels, some of the light is reflected back onto a photodetector. The intensity of light reflected back captures the amount of blood in your blood vessel via a waveform.

Depending on the color of light used, different biometrics can be captured through their unique waveforms. For example, when your blood vessels are expanded, more green light will be absorbed compared to when they're contracted. This phenomenon allows your pulse to be accurately taken.

To gather an accurate pulse reading, you must remain relatively still for Live Heart Rate to take its full 10-second measurement. If Oura detects too much motion, your Live Heart Rate won't be available.

(<https://support.ouraring.com/hc/en-us/articles/4410651298963-Live-Heart-Rate>)

How Oura Measures Resting Heart Rate

Oura monitors your RHR by detecting changes in your blood pulse volume through infrared PPG sensors found inside your ring.

Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
- **Lowest RHR:** Your lowest RHR value captured during the night

Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

(<https://support.ouraring.com/hc/en-us/articles/360025588793-Resting-Heart-Rate>)

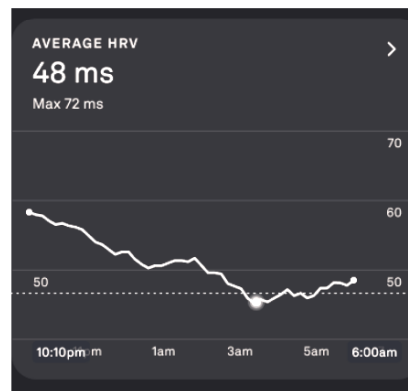
How Oura Calculates HRV

Oura provides a reliable and accurate measurement of your autonomic nervous system's activity by calculating your nighttime HRV.

Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV value you see is the mean of all five-minute samples taken while you sleep.

Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
- **Max HRV:** Your highest HRV value recorded during the night



You can monitor your HRV under the **Readiness** or **Trends** views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

(https://support.ouraring.com/hc/en-us/articles/360025441974-Heart-Rate-Variability#h_01GVP1CDV0Y6YKY24R7TK5X5G1)

How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your [nighttime PPG data](#) – the same signal it uses to measure your [resting heart rate](#) and [heart rate variability](#).

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., [caffeine](#), [emotion](#), [exercise](#)), and it's even been shown that your rate increases when you know it is being counted.



You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

- When you breathe in, your heart rate increases.
- When you breathe out, your heart rate slows down.

This helps your body take in oxygen and expel CO₂ with as few heartbeats as possible. Because this pattern is predictable, you can measure the variation between heartbeats to determine your respiratory rate.

https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B_

Software

Oura Ring 4 with active membership includes:

- Detailed sleep analysis
- 24/7 heart rate tracking (Daytime, Nighttime, Activity)
- Integration with third-party health and wellness apps like Strava
- Advanced temperature monitoring
- Blood Oxygen Sensing (SpO2)
- Daytime Stress
- Dynamic activity goals with Automatic Activity Detection
- Heart Health (Cardiovascular Age, Cardio Capacity (VO2 Max))
- Metabolic Health (Glucose Tracking, Meals)
- Oura Labs
- Resilience
- Weekly, monthly, quarterly, yearly, and anniversary reports
- Women's Health (Cycle Insights, Pregnancy Insights)

Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

1[c] a controller configured to:

The Oura Accused Product includes a controller. For example, the Oura Accused Product comprises an Infineon PSoC 6 Controller.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

<https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf>

	U.S. Patent No. 10,642,359
1[c][i] obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor,	The Oura Accused Product obtains a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor. For example, the Oura Accused Product obtains a movement signal of an object, such as a human or human body part, using a motion sensor that measures motion, such as an accelerometer. The Oura Accused Product also obtains a biological signal of the object, such as blood oxygen (SpO₂) levels, heart rate, heart rate variability, and/or respiration rate, using the biological sensor. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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	U.S. Patent No. 10,642,359
	Live Heart Rate The latest reading of your pulse is displayed as beats per minute (BPM) beneath the heart rate graph, with a time indication of how long ago the last update was measured. Tapping the heart next to this number will update it with your current pulse measurement. Check out the Live Heart Rate article for more information. Restorative Time Restorative Time marks moments during the day when your body falls into a restful recovery state. Tracking these moments is a good way to measure your readiness. For more information on getting the most out of this feature, see our Restorative Time article. Stress Moments where your body has felt high levels of physiological stress are marked in pink on the heart rate graph. For more information, see the Daytime Stress article. Daytime Heart Rate Measurement Oura takes measurements of your daytime heart rate for one full minute, every five minutes, using the Oura Ring's green LEDs. To preserve battery and maximize accuracy, a daytime heart rate measurement is only taken under optimal conditions, which include low movement and balanced average body temperature. Due to prioritizing these conditions, you <i>may</i> not receive an automatically updated heart rate for up to 30 minutes, but it can be manually updated at any time by using Live Heart Rate. Also note that "low movement" generally includes everyday actions, up to excessive motion associated with workouts. Activity Heart Rate The Activity Heart Rate feature is used to record your exercises. Heart rate measurements taken from physical activity are highlighted in orange on the graph. https://support.ouraring.com/hc/en-us/articles/4410656562579-Heart-Rate-Graph

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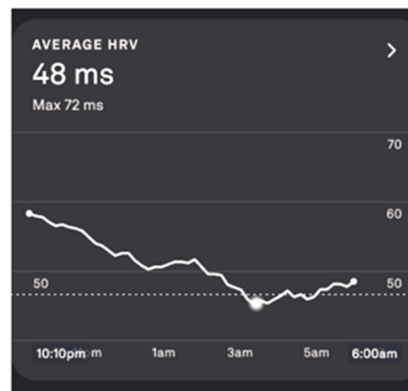
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https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B_

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Oura Ring 4 with active membership includes:

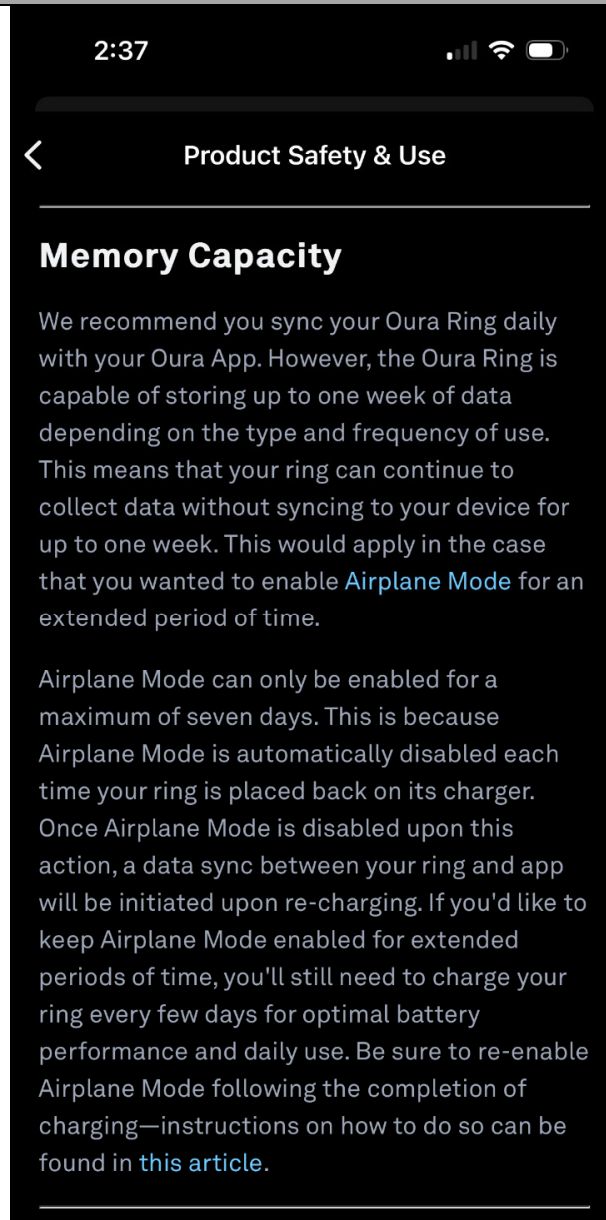
- Detailed sleep analysis
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(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

	U.S. Patent No. 10,642,359
1[c][ii] determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, and	The Oura Accused Product determines, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal. For example, the Oura Accused Product determines, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, such as computing, storing, and/or transmitting blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate data.



(Oura App → Help → Product Safety & Use → Memory Capacity)

Memory Capacity

We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable **Airplane Mode** for an extended period of time.

Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in **this article**.

(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
	Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring. The Oura App is available on the App Store and Google Play. The Oura App is supported on iOS devices running Apple iOS 15 and higher, or Android devices running Android 8.0 and higher with Google Play services. Your device must support Bluetooth 4.0. Ring firmware updates are accessible directly from your Oura App. (https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

At Oura, we're committed to innovating and improving our health sensing algorithms through rigorous scientific processes, as well as sharing the research and validation with our community. Oura's Sleep Staging Algorithm 2.0, which matches performance with the gold standard [polysomnography \(PSG\) sleep lab test](#).

The result of more than two years of extensive research and development, this updated algorithm is based on a 2021 research study published in [Sensors](#) (a leading international, peer-reviewed, and open-access journal on the science and technology of sensors), which showcases the results of the algorithm.

How Accurately Does Oura Measure Sleep?

By leveraging every signal from the Oura Ring (movement, temperature, heart rate, and HRV), the *Sensors* study found Oura's sleep staging algorithm achieves **79% agreement** with gold-standard polysomnography (PSG) for classifying the [four stages of sleep](#) (wake, light, deep, and rapid eye movement (REM) sleep). By contrast, most commercial wrist-based wearables are limited to [60-65% agreement](#).

Oura's new sleep staging algorithm was developed using advanced machine learning techniques and one of the largest wearable sleep datasets collected to date. This one-of-a-kind dataset includes diverse participants across multiple continents and sleep laboratories, demonstrating the power of using all of the signals of the Oura Ring to maximize accuracy.

Read [the full paper in Sensors](#) and see key takeaways below.
(<https://ouraring.com/blog/new-sleep-staging-algorithm/>)

Sensors

- Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life
- Red and infrared LEDs measure blood oxygen while you sleep
- Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep
- Digital temperature sensor measures average body temperature variations
- Accelerometer tracks movement and activity 24/7

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(<https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2>)

What Is Live Heart Rate?

With Live Heart Rate, you can check your heart rate at any time during the day. Just tap to see your heart rate instantly. You can access Live Heart Rate through the 24-hour [heart rate graph](#).

Oura uses green LEDs to measure your heart rate in real-time. The process uses photoplethysmography (PPG) technology at a frequency of 50Hz.

PPG technology works by shining a light onto the surface of your skin. When this reaches your blood vessels, some of the light is reflected back onto a photodetector. The intensity of light reflected back captures the amount of blood in your blood vessel via a waveform.

Depending on the color of light used, different biometrics can be captured through their unique waveforms. For example, when your blood vessels are expanded, more green light will be absorbed compared to when they're contracted. This phenomenon allows your pulse to be accurately taken.

To gather an accurate pulse reading, you must remain relatively still for Live Heart Rate to take its full 10-second measurement. If Oura detects too much motion, your Live Heart Rate won't be available.

(<https://support.ouraring.com/hc/en-us/articles/4410651298963-Live-Heart-Rate>)

How Oura Measures Resting Heart Rate

Oura monitors your RHR by detecting changes in your blood pulse volume through infrared PPG sensors found inside your ring.

Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
- **Lowest RHR:** Your lowest RHR value captured during the night

Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

(<https://support.ouraring.com/hc/en-us/articles/360025588793-Resting-Heart-Rate>)

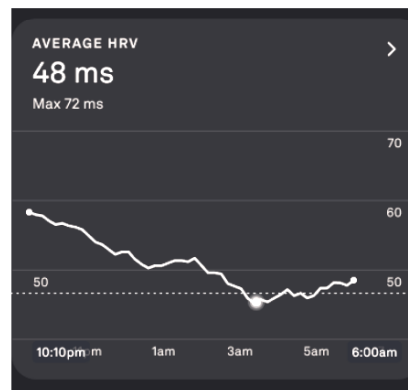
How Oura Calculates HRV

Oura provides a reliable and accurate measurement of your autonomic nervous system's activity by calculating your nighttime HRV.

Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV value you see is the mean of all five-minute samples taken while you sleep.

Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
- **Max HRV:** Your highest HRV value recorded during the night



You can monitor your HRV under the **Readiness** or **Trends** views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

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How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your [nighttime PPG data](#) – the same signal it uses to measure your [resting heart rate](#) and [heart rate variability](#).

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., [caffeine](#), [emotion](#), [exercise](#)), and it's even been shown that your rate increases when you know it is being counted.



You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

- When you breathe in, your heart rate increases.
- When you breathe out, your heart rate slows down.

This helps your body take in oxygen and expel CO₂ with as few heartbeats as possible. Because this pattern is predictable, you can measure the variation between heartbeats to determine your respiratory rate.

https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B_

Software

Oura Ring 4 with active membership includes:

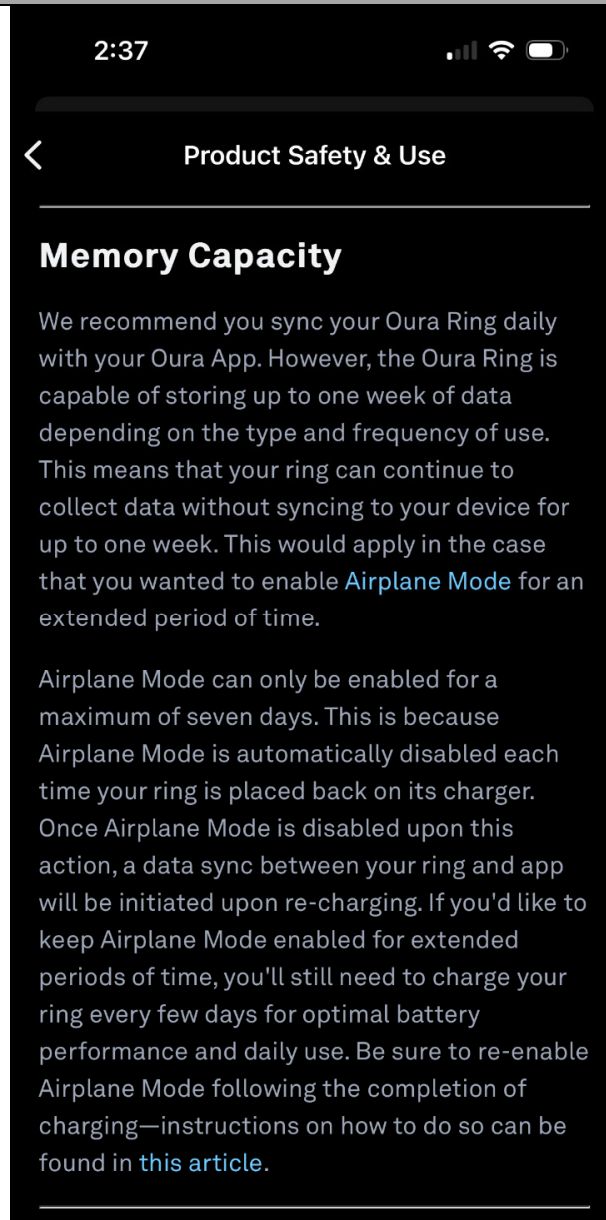
- Detailed sleep analysis
- 24/7 heart rate tracking (Daytime, Nighttime, Activity)
- Integration with third-party health and wellness apps like Strava
- Advanced temperature monitoring
- Blood Oxygen Sensing (SpO2)
- Daytime Stress
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- Resilience
- Weekly, monthly, quarterly, yearly, and anniversary reports
- Women's Health (Cycle Insights, Pregnancy Insights)

Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

U.S. Patent No. 10,642,359	
1[c][iii] perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition.	The Oura Accused Product performs the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition. For example, the Oura Accused Product performs the specified function of computing, storing, and/or transmitting sleep-related data or data collected while the user is determined to be asleep using a select portion of the biological signal, such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate, obtained during a select period that corresponds to a specified condition, such as the user sleeping or a movement signal indicative of the user sleeping.



(Oura App → Help → Product Safety & Use → Memory Capacity)

Memory Capacity

We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable **Airplane Mode** for an extended period of time.

Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in **this article**.

(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
	Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring. The Oura App is available on the App Store and Google Play. The Oura App is supported on iOS devices running Apple iOS 15 and higher, or Android devices running Android 8.0 and higher with Google Play services. Your device must support Bluetooth 4.0. Ring firmware updates are accessible directly from your Oura App. (https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

At Oura, we're committed to innovating and improving our health sensing algorithms through rigorous scientific processes, as well as sharing the research and validation with our community. Oura's Sleep Staging Algorithm 2.0, which matches performance with the gold standard [polysomnography \(PSG\) sleep lab test](#).

The result of more than two years of extensive research and development, this updated algorithm is based on a 2021 research study published in [Sensors](#) (a leading international, peer-reviewed, and open-access journal on the science and technology of sensors), which showcases the results of the algorithm.

How Accurately Does Oura Measure Sleep?

By leveraging every signal from the Oura Ring (movement, temperature, heart rate, and HRV), the *Sensors* study found Oura's sleep staging algorithm achieves **79% agreement** with gold-standard polysomnography (PSG) for classifying the [four stages of sleep](#) (wake, light, deep, and rapid eye movement (REM) sleep). By contrast, most commercial wrist-based wearables are limited to [60-65% agreement](#).

Oura's new sleep staging algorithm was developed using advanced machine learning techniques and one of the largest wearable sleep datasets collected to date. This one-of-a-kind dataset includes diverse participants across multiple continents and sleep laboratories, demonstrating the power of using all of the signals of the Oura Ring to maximize accuracy.

Read [the full paper in Sensors](#) and see key takeaways below.
(<https://ouraring.com/blog/new-sleep-staging-algorithm/>)

How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

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Oura provides you with the following RHR metrics:

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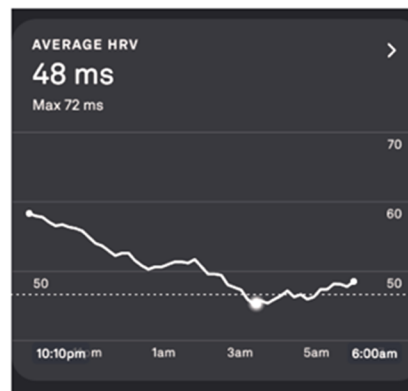
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Author: Oura Team | February 6, 2024



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(https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B)

Software

Oura Ring 4 with active membership includes:

- Detailed sleep analysis
- 24/7 heart rate tracking (Daytime, Nighttime, Activity)
- Integration with third-party health and wellness apps like Strava
- Advanced temperature monitoring
- Blood Oxygen Sensing (SpO2)
- Daytime Stress
- Dynamic activity goals with Automatic Activity Detection
- Heart Health (Cardiovascular Age, Cardio Capacity (VO2 Max))
- Metabolic Health (Glucose Tracking, Meals)
- Oura Labs
- Resilience
- Weekly, monthly, quarterly, yearly, and anniversary reports
- Women's Health (Cycle Insights, Pregnancy Insights)

Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

	U.S. Patent No. 10,642,359
Independent Claim 7	
7[pre] 7. A wearable device comprising:	To the extent that the preamble is limiting, the Oura Accused Product is a wearable device.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

[\(https://ouraring.com/blog/oura-ring-4/\)](https://ouraring.com/blog/oura-ring-4/)

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](#) and [Oura Ring Generation 3](#).
(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
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U.S. Patent No. 10,642,359	
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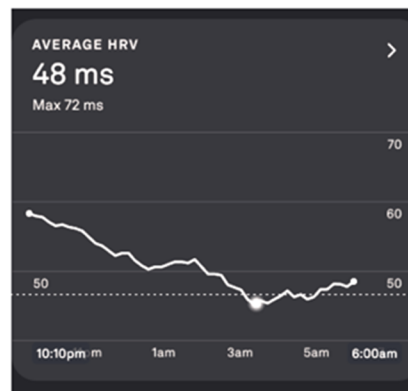
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This helps your body take in oxygen and expel CO₂ with as few heartbeats as possible. Because this pattern is predictable, you can measure the variation between heartbeats to determine your respiratory rate.

(https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B)

Software

Oura Ring 4 with active membership includes:

- Detailed sleep analysis
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- Blood Oxygen Sensing (SpO2)
- Daytime Stress
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- Heart Health (Cardiovascular Age, Cardio Capacity (VO2 Max))
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- Resilience
- Weekly, monthly, quarterly, yearly, and anniversary reports
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Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4>)

7[c] a controller configured to:

The Oura Accused Product includes a controller. For example, the Oura Accused Product comprises an Infineon PSoC 6 Controller.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

<https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf>

	U.S. Patent No. 10,642,359
7[c][i] obtain a movement signal of an object external to the wearable device using the motion sensor;	The Oura Accused Product obtains a movement signal of an object external to the wearable device using the motion sensor. For example, the Oura Accused Product obtains a movement signal of an object external to the wearable device, such as a human user or human user's body part, using the motion sensor that measures motion, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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7[c][ii] obtain a biological signal of the object using the biological sensor;	The Oura Accused Product obtains a biological signal of the object using the biological sensor. For example, the Oura Accused Product obtains a biological signal of the object, such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate, using the biological sensor, such as a red and/or infrared (IR) LED and sensor or a green and/or infrared (IR) LED and sensor.

U.S. Patent No. 10,642,359	
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How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

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Live Heart Rate

The latest reading of your pulse is displayed as beats per minute (BPM) beneath the heart rate graph, with a time indication of how long ago the last update was measured. Tapping the heart next to this number will update it with your current pulse measurement. Check out [the Live Heart Rate article](#) for more information.

Restorative Time

Restorative Time marks moments during the day when your body falls into a restful recovery state. Tracking these moments is a good way to measure your readiness. For more information on getting the most out of this feature, see [our Restorative Time article](#).

Stress

Moments where your body has felt high levels of physiological stress are marked in pink on the heart rate graph. For more information, see the [Daytime Stress article](#).

Daytime Heart Rate Measurement

Oura takes measurements of your daytime heart rate for one full minute, every five minutes, using the Oura Ring's green LEDs. To preserve battery and maximize accuracy, a daytime heart rate measurement is only taken under optimal conditions, which include low movement and balanced average body temperature.

Due to prioritizing these conditions, you *may* not receive an automatically updated heart rate for up to 30 minutes, but it can be manually updated at any time by using [Live Heart Rate](#). Also note that "low movement" generally includes everyday actions, up to excessive motion associated with workouts.

Activity Heart Rate

The [Activity Heart Rate feature](#) is used to record your exercises. Heart rate measurements taken from physical activity are highlighted in orange on the graph.

(<https://support.ouraring.com/hc/en-us/articles/4410656562579-Heart-Rate-Graph>)

How Oura Measures Resting Heart Rate

Oura monitors your RHR by detecting changes in your blood pulse volume through infrared PPG sensors found inside your ring.

Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
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Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

(<https://support.ouraring.com/hc/en-us/articles/360025588793-Resting-Heart-Rate>)

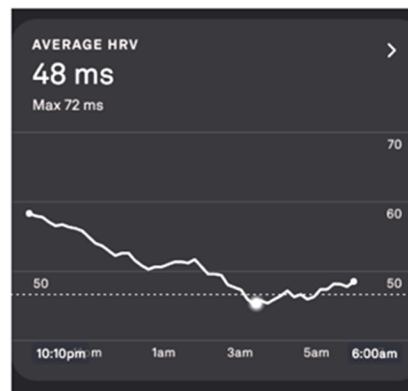
How Oura Calculates HRV

Oura provides a reliable and accurate measurement of your autonomic nervous system's activity by calculating your nighttime HRV.

Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV value you see is the mean of all five-minute samples taken while you sleep.

Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
- **Max HRV:** Your highest HRV value recorded during the night



You can monitor your HRV under the **Readiness** or **Trends** views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

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How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your [nighttime PPG data](#) – the same signal it uses to measure your [resting heart rate](#) and [heart rate variability](#).

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., [caffeine](#), [emotion](#), [exercise](#)), and it's even been shown that your rate increases when you know it is being counted.



You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

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Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

	U.S. Patent No. 10,642,359
7[c][iii] identify a period, of the movement signal, the movement signal is determined to be less than a specified condition; and	The Oura Accused Product identifies a period, of the movement signal, the movement signal is determined to be less than a specified condition. For example, the Oura Accused Product identifies a period, of the movement signal, where the movement signal is determined to be less than a specified condition, such as movement level condition below which is indicative of the user sleeping.

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

At Oura, we're committed to innovating and improving our health sensing algorithms through rigorous scientific processes, as well as sharing the research and validation with our community. Oura's Sleep Staging Algorithm 2.0, which matches performance with the gold standard [polysomnography \(PSG\) sleep lab test](#).

The result of more than two years of extensive research and development, this updated algorithm is based on a 2021 research study published in [Sensors](#) (a leading international, peer-reviewed, and open-access journal on the science and technology of sensors), which showcases the results of the algorithm.

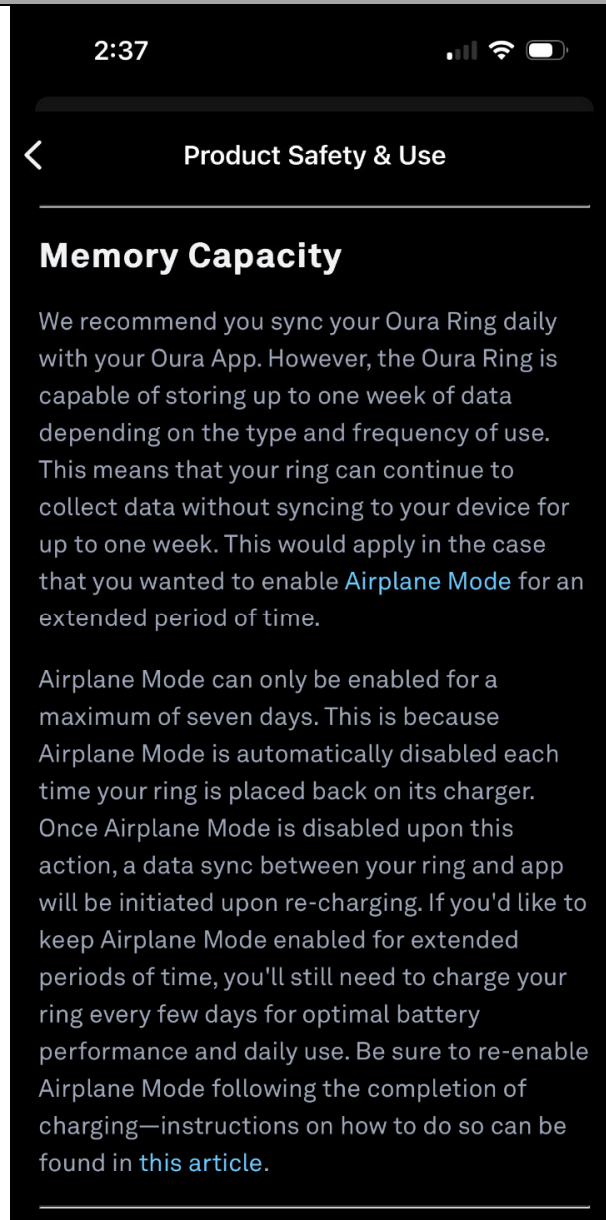
How Accurately Does Oura Measure Sleep?

By leveraging every signal from the Oura Ring (movement, temperature, heart rate, and HRV), the *Sensors* study found Oura's sleep staging algorithm achieves **79% agreement** with gold-standard polysomnography (PSG) for classifying the [four stages of sleep](#) (wake, light, deep, and rapid eye movement (REM) sleep). By contrast, most commercial wrist-based wearables are limited to [60-65% agreement](#).

Oura's new sleep staging algorithm was developed using advanced machine learning techniques and one of the largest wearable sleep datasets collected to date. This one-of-a-kind dataset includes diverse participants across multiple continents and sleep laboratories, demonstrating the power of using all of the signals of the Oura Ring to maximize accuracy.

Read [the full paper in Sensors](#) and see key takeaways below.
(<https://ouraring.com/blog/new-sleep-staging-algorithm/>)

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7[c][iv] perform a specified function using a select portion of the biological signal corresponding to the period.	The Oura Accused Product performs a specified function using a select portion of the biological signal corresponding to the period. For example, the Oura Accused Product performs a specified function, such as computing, storing, and/or transmitting sleep-related data or data collected while the user is determined to be asleep, using a select portion of the biological signal, such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate, corresponding to the period when the user's movement level is indicative of sleeping.



(Oura App → Help → Product Safety & Use → Memory Capacity)

Memory Capacity

We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable **Airplane Mode** for an extended period of time.

Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in **this article**.

(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
	Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring. The Oura App is available on the App Store and Google Play. The Oura App is supported on iOS devices running Apple iOS 15 and higher, or Android devices running Android 8.0 and higher with Google Play services. Your device must support Bluetooth 4.0. Ring firmware updates are accessible directly from your Oura App. (https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

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Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

(<https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2>)

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Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
- **Lowest RHR:** Your lowest RHR value captured during the night

Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

(<https://support.ouraring.com/hc/en-us/articles/360025588793-Resting-Heart-Rate>)

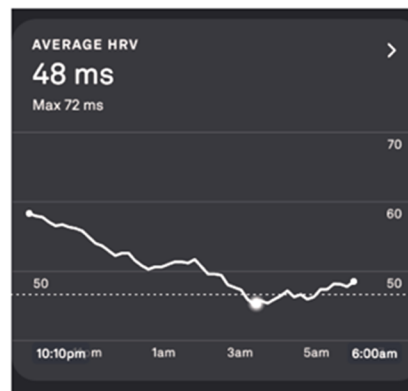
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Author: Oura Team | February 6, 2024



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	U.S. Patent No. 10,642,359
Independent Claim 14	
14[pre] 14. A wearable device comprising:	To the extent that the preamble is limiting, the Oura Accused Product is a wearable device.

A Wearable That's Truly Wearable, Day and Night



Oura Ring 4 is sleeker and more comfortable than its predecessors, allowing it to blend seamlessly into your lifestyle. With a fully titanium coating inside and out, it's gentle to the touch, even on sensitive skin, and slips on and off your finger with ease.

Plus, the powerful health sensors are now recessed, reducing friction and pressure on your fingers when shaking hands or gripping objects. With the lower profile and advancements in comfort of Oura Ring 4, you might even forget you're wearing it—until the compliments start coming.

[\(https://ouraring.com/blog/oura-ring-4/\)](https://ouraring.com/blog/oura-ring-4/)

Hardware

Oura Ring 4 features a fully titanium design and recessed sensors on the interior, enhancing both durability and wearability. Oura Ring 4 is available in six finishes: Black, Silver, Stealth, Brushed Silver, Gold, and Rose Gold.

Oura Ring 4 Ceramic is made from high-performance zirconia ceramic, a premium material selected for its scratch resistance, luster, and color. Oura Ring 4 Ceramic is available in four colors: Midnight, Cloud, Petal, and Tide.

All Oura Ring 4 models are available in sizes 4-15.

The sizing for Oura Ring 4 is slightly different from Oura Ring Gen3. Make sure to use the right sizing kit for the most accurate fit.

- Oura Ring 4 Sizing Kit: white ring sizers, sizes 4–15
- Oura Ring Gen3 Sizing Kit: black ring sizers, sizes 6–13

For the best fit and most accurate readings, align your ring so that the sensors are against the palm side of your finger.

<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>

General Use

You can wear your Oura Ring 24/7. It's built for all-day comfort.

We don't recommend wearing your Oura Ring adjacent to other rings as this may cause discomfort or unintended scratching.

We recommend wearing your Oura Ring on your non-dominant hand. By doing so, you'll limit friction (and scratches to your ring) with items and surfaces you grab and touch each day.

The Oura Ring is water-resistant up to 100 meters (~330 feet), meaning that you can wear it while showering, swimming, or partaking in recreational diving (e.g. *snorkeling*), but we don't recommend using your ring while scuba diving or keeping it submerged underwater for over 12 hours.

Oura Ring operating temperature ranges from -10–52°C / 14–125°F. You can safely wear your ring in the shower, hot tubs, saunas, ice baths, and cryotherapy tanks. However, extended exposure to extreme temperatures (below -20°C / -4°F or above 60°C / 140°F) may lead to battery damage.

For more product specifications, see [Oura Ring 4](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ) and [Oura Ring Generation 3](https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ).
(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
14[a] a motion sensor;	The Oura Accused Product includes a motion sensor. For example, the Oura Accused Product includes a motion sensor that measures motion, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

U.S. Patent No. 10,642,359	
	Sensors • Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)
14[b] a biological sensor; and	The Oura Accused Product includes a biological sensor. For example, the Oura Accused Product includes a biological sensor that measures biological information, such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate.

U.S. Patent No. 10,642,359	
	Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4

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How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

(<https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2>)

	U.S. Patent No. 10,642,359
	Live Heart Rate The latest reading of your pulse is displayed as beats per minute (BPM) beneath the heart rate graph, with a time indication of how long ago the last update was measured. Tapping the heart next to this number will update it with your current pulse measurement. Check out the Live Heart Rate article for more information. Restorative Time Restorative Time marks moments during the day when your body falls into a restful recovery state. Tracking these moments is a good way to measure your readiness. For more information on getting the most out of this feature, see our Restorative Time article. Stress Moments where your body has felt high levels of physiological stress are marked in pink on the heart rate graph. For more information, see the Daytime Stress article. Daytime Heart Rate Measurement Oura takes measurements of your daytime heart rate for one full minute, every five minutes, using the Oura Ring's green LEDs. To preserve battery and maximize accuracy, a daytime heart rate measurement is only taken under optimal conditions, which include low movement and balanced average body temperature. Due to prioritizing these conditions, you <i>may</i> not receive an automatically updated heart rate for up to 30 minutes, but it can be manually updated at any time by using Live Heart Rate. Also note that "low movement" generally includes everyday actions, up to excessive motion associated with workouts. Activity Heart Rate The Activity Heart Rate feature is used to record your exercises. Heart rate measurements taken from physical activity are highlighted in orange on the graph. https://support.ouraring.com/hc/en-us/articles/4410656562579-Heart-Rate-Graph

How Oura Measures Resting Heart Rate

Oura monitors your RHR by detecting changes in your blood pulse volume through infrared PPG sensors found inside your ring.

Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
- **Lowest RHR:** Your lowest RHR value captured during the night

Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

(<https://support.ouraring.com/hc/en-us/articles/360025588793-Resting-Heart-Rate>)

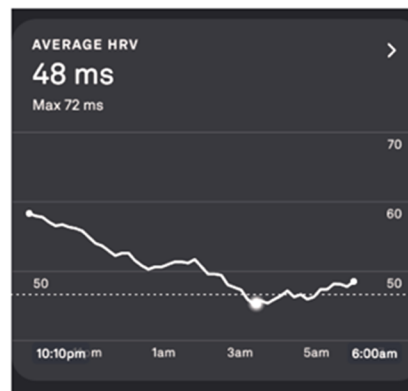
How Oura Calculates HRV

Oura provides a reliable and accurate measurement of your autonomic nervous system's activity by calculating your nighttime HRV.

Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV value you see is the mean of all five-minute samples taken while you sleep.

Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
- **Max HRV:** Your highest HRV value recorded during the night



You can monitor your HRV under the **Readiness** or **Trends** views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

(https://support.ouraring.com/hc/en-us/articles/360025441974-Heart-Rate-Variability#h_01GVP1CDV0Y6YKY24R7TK5X5G1)

How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your [nighttime PPG data](#) – the same signal it uses to measure your [resting heart rate](#) and [heart rate variability](#).

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., [caffeine](#), [emotion](#), [exercise](#)), and it's even been shown that your rate increases when you know it is being counted.



You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

- When you breathe in, your heart rate increases.
- When you breathe out, your heart rate slows down.

This helps your body take in oxygen and expel CO₂ with as few heartbeats as possible. Because this pattern is predictable, you can measure the variation between heartbeats to determine your respiratory rate.

(https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B)

Software

Oura Ring 4 with active membership includes:

- Detailed sleep analysis
- 24/7 heart rate tracking (Daytime, Nighttime, Activity)
- Integration with third-party health and wellness apps like Strava
- Advanced temperature monitoring
- Blood Oxygen Sensing (SpO2)
- Daytime Stress
- Dynamic activity goals with Automatic Activity Detection
- Heart Health (Cardiovascular Age, Cardio Capacity (VO2 Max))
- Metabolic Health (Glucose Tracking, Meals)
- Oura Labs
- Resilience
- Weekly, monthly, quarterly, yearly, and anniversary reports
- Women's Health (Cycle Insights, Pregnancy Insights)

Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4>)

14[c] a controller configured to:

The Oura Accused Product includes a controller. For example, the Oura Accused Product comprises an Infineon PSoC 6 Controller.

UNMATCHED SOLUTIONS FOR THE INTERNET OF THINGS

The Oura Ring

OURA

The Oura Ring is the world's most-advanced wearable that helps you achieve restorative sleep. This innovation is delivered through PSoC 6 MCUs, enabling the following features:

SMALL FORM FACTOR

Integration of several functions into a single PSoC 6 MCU helped reduce the size of the ring over the prior generation by 50%.

DATA STORAGE

PSoC 6's on-chip flash memory enables storage of up to six weeks of user data.

BLUETOOTH LOW ENERGY

The built-in BLE wireless connectivity in PSoC 6 provides communication to the mobile app device.



EXTENDED BATTERY LIFE

PSoC 6's ultra-low power was able to extend battery life to seven days, 3X longer than the prior generation ring.

SENSOR FUSION

PSoC 6, as the host processor, connects different biometric sensors in the ring to collect and process data.

DATA PROCESSING

The dual Cortex-M4 and Cortex-M0+ cores in PSoC 6 enabled 10X more processing capability vs. the prior-generation ring.

(<https://www.infineon.com/assets/row/public/documents/30/39/infineon-psoc-6-mcu-the-new-standard-for-the-internet-of-things-productbrochure-en.pdf>)

	U.S. Patent No. 10,642,359
14[c][i] obtain a movement signal of an object external to the wearable device using the motion sensor;	The Oura Accused Product obtains a movement signal of an object external to the wearable device using the motion sensor. For example, the Oura Accused Product obtains a movement signal of an object external to the wearable device, such as a human user or human user's body part, using the motion sensor that measures motion, such as an accelerometer. Sensors • Smart Sensing is a sensing platform designed to give you the most complete picture of your health. The research-grade sensors work with an algorithm that adapts to the unique structure and skin tone of your finger and your current activity to deliver the most accurate, continuous data, day and night—all while optimizing battery life • Red and infrared LEDs measure blood oxygen while you sleep • Green and infrared PPG sensors alternate to measure heart rate 24/7 and heart rate variability and respiration rate during your sleep • Digital temperature sensor measures average body temperature variations • Accelerometer tracks movement and activity 24/7 (https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4)

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14[c][ii] obtain a biological signal of the object using the biological sensor;	The Oura Accused Product obtains a biological signal of the object using the biological sensor. For example, the Oura Accused Product obtains a biological signal of the object, such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate, using the biological sensor, such as a red and/or infrared (IR) LED and sensor and/or a green or infrared (IR) LED and sensor.

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Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
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Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

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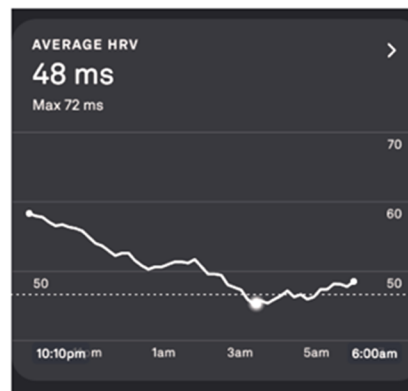
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Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
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You can monitor your HRV under the **Readiness** or **Trends** views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

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How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your [nighttime PPG data](#) – the same signal it uses to measure your [resting heart rate](#) and [heart rate variability](#).

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., [caffeine](#), [emotion](#), [exercise](#)), and it's even been shown that your rate increases when you know it is being counted.



You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

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https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B_

Software

Oura Ring 4 with active membership includes:

- Detailed sleep analysis
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Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4>)

	U.S. Patent No. 10,642,359
14[c][iii] determine a select period, of the movement signal, that meets a specified condition;	The Oura Accused Product determines a select period, of the movement signal, that meets a specified condition. For example, the Oura Accused Product determines a select period, of the movement signal, that meets a specified condition, such as the user sleeping or a movement signal condition indicative of the user sleeping.

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

At Oura, we're committed to innovating and improving our health sensing algorithms through rigorous scientific processes, as well as sharing the research and validation with our community. Oura's Sleep Staging Algorithm 2.0, which matches performance with the gold standard [polysomnography \(PSG\) sleep lab test](#).

The result of more than two years of extensive research and development, this updated algorithm is based on a 2021 research study published in [Sensors](#) (a leading international, peer-reviewed, and open-access journal on the science and technology of sensors), which showcases the results of the algorithm.

How Accurately Does Oura Measure Sleep?

By leveraging every signal from the Oura Ring (movement, temperature, heart rate, and HRV), the *Sensors* study found Oura's sleep staging algorithm achieves **79% agreement** with gold-standard polysomnography (PSG) for classifying the [four stages of sleep](#) (wake, light, deep, and rapid eye movement (REM) sleep). By contrast, most commercial wrist-based wearables are limited to [60-65% agreement](#).

Oura's new sleep staging algorithm was developed using advanced machine learning techniques and one of the largest wearable sleep datasets collected to date. This one-of-a-kind dataset includes diverse participants across multiple continents and sleep laboratories, demonstrating the power of using all of the signals of the Oura Ring to maximize accuracy.

Read [the full paper in Sensors](#) and see key takeaways below.
(<https://ouraring.com/blog/new-sleep-staging-algorithm/>)

	U.S. Patent No. 10,642,359
14[c][iv] determine, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined select period; and	The Oura Accused Product determines, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined select period. For example, the Oura Accused Product determines, as a valid signal, at least a portion of the biometric signal, such as blood oxygen (SpO ₂) levels, heart rate, heart rate variability, and/or respiration rate, that is determined to correspond to the determined select period when the user's movement level is indicative of sleeping.

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Sensors

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U.S. Patent No. 10,642,359			
	Sensors	• Red and infrared LEDs measure blood oxygen levels • Green and infrared LEDs alternate to measure heart rate and heart rate variability 24/7 and respiration rate • Digital sensor measures temperature trends • Accelerometer tracks movement and activity 24/7	
			https://support.ouraring.com/hc/en-us/articles/33045011508115-Oura-Ring-4

How Does Blood Oxygen Sensing (SpO2) Work?

The Oura Ring measures peripheral oxygen saturation (SpO2) levels in your blood by sending light from the ring's red and infrared LEDs into your finger while you sleep, and then measuring how much light is reflected. Richly oxygenated blood reflects more red light than infrared light, while poorly oxygenated blood reflects more infrared light than red light.

Your blood oxygen measurements are located on the Oura App's Sleep screen. The **Average Blood Oxygen** reading is the percentage of your blood that contains oxygen. The **Breathing Regularity** graph displays variations in how much oxygen is present in your blood. This is different from **respiratory rate**, which measures how many breaths per minute (BPM) you take. Average Blood Oxygen and Breathing Regularity are measured during **sleep periods longer than three hours**, but the Oura App only displays the results for the longest sleep period of the day.

Please note that you may see the red LEDs light up whenever your Oura Ring detects that you may be asleep (based on your heart rate, movement, and respiratory rate), including during the day.

(<https://support.ouraring.com/hc/en-us/articles/7328398760851-Blood-Oxygen-Sensing-SpO2>)

Live Heart Rate

The latest reading of your pulse is displayed as beats per minute (BPM) beneath the heart rate graph, with a time indication of how long ago the last update was measured. Tapping the heart next to this number will update it with your current pulse measurement. Check out [the Live Heart Rate article](#) for more information.

Restorative Time

Restorative Time marks moments during the day when your body falls into a restful recovery state. Tracking these moments is a good way to measure your readiness. For more information on getting the most out of this feature, see [our Restorative Time article](#).

Stress

Moments where your body has felt high levels of physiological stress are marked in pink on the heart rate graph. For more information, see the [Daytime Stress article](#).

Daytime Heart Rate Measurement

Oura takes measurements of your daytime heart rate for one full minute, every five minutes, using the Oura Ring's green LEDs. To preserve battery and maximize accuracy, a daytime heart rate measurement is only taken under optimal conditions, which include low movement and balanced average body temperature.

Due to prioritizing these conditions, you *may* not receive an automatically updated heart rate for up to 30 minutes, but it can be manually updated at any time by using [Live Heart Rate](#). Also note that "low movement" generally includes everyday actions, up to excessive motion associated with workouts.

Activity Heart Rate

The [Activity Heart Rate feature](#) is used to record your exercises. Heart rate measurements taken from physical activity are highlighted in orange on the graph.

(<https://support.ouraring.com/hc/en-us/articles/4410656562579-Heart-Rate-Graph>)

How Oura Measures Resting Heart Rate

Oura monitors your RHR by detecting changes in your blood pulse volume through infrared PPG sensors found inside your ring.

Each time that your heart beats, blood is pumped out to the arteries located in your hands and fingers. The PPG sensors are able to detect these changes in blood flow and volume using light reflection. Each pulse causes the arteries in your finger to alternate between swelling and contracting. By shining a light on your skin, changes in light reflected back from the wavering volume of red blood cells in your arteries are accounted for. From here, PPG can represent these blood flow changes through a visual waveform that represents the activity of your heart.

Oura provides you with the following RHR metrics:

- **Average RHR:** Your average RHR captured during the night
- **Lowest RHR:** Your lowest RHR value captured during the night

Your RHR values are captured every 10 minutes throughout the night. You can find your RHR in your Readiness and Sleep detail views and in **Trends**.

(<https://support.ouraring.com/hc/en-us/articles/360025588793-Resting-Heart-Rate>)

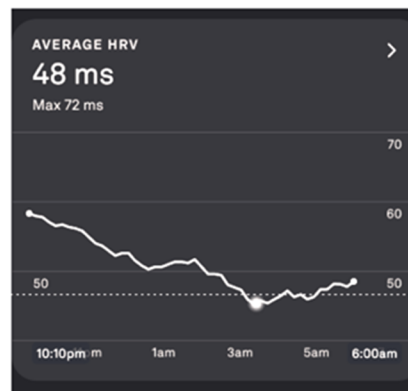
How Oura Calculates HRV

Oura provides a reliable and accurate measurement of your autonomic nervous system's activity by calculating your nighttime HRV.

Oura takes five-minute samples, not just a single reading, throughout the night. This continuous tracking offers greater accuracy and a more detailed view of your body's response to stress and preparation for the day ahead. The average HRV value you see is the mean of all five-minute samples taken while you sleep.

Key HRV metrics:

- **Average HRV:** Your average HRV measured over the entire night
- **Max HRV:** Your highest HRV value recorded during the night



You can monitor your HRV under the **Readiness** or **Trends** views. Tracking HRV in the Trends view is particularly helpful for assessing your recovery over the past week and identifying any internal or external stressors that may require attention.

(https://support.ouraring.com/hc/en-us/articles/360025441974-Heart-Rate-Variability#h_01GVP1CDV0Y6YKY24R7TK5X5G1)

How Oura Measures Respiratory Rate From Your Finger



Author: Oura Team | February 6, 2024



Respiratory rate can be measured directly by counting the rise and fall of someone's chest or indirectly by reading it from another signal in your body, like your heart's activity. A widely used method is deriving respiration from photoplethysmography (PPG), the same light pulses used to measure heart rate in Oura and many hospital devices.

How Does It Work?

Oura derives your respiratory rate exclusively from your [nighttime PPG data](#) – the same signal it uses to measure your [resting heart rate](#) and [heart rate variability](#).

Nighttime is an ideal time to measure respiration because your body is in a consistent state. During the day, many factors can influence your respiratory rate (e.g., [caffeine](#), [emotion](#), [exercise](#)), and it's even been shown that your rate increases when you know it is being counted.



You can measure respiration in your heartbeat because your heart and lungs are closely connected by a process called respiratory sinus arrhythmia. Although it sounds serious, it's a completely normal and simple partnership:

- When you breathe in, your heart rate increases.
- When you breathe out, your heart rate slows down.

This helps your body take in oxygen and expel CO₂ with as few heartbeats as possible. Because this pattern is predictable, you can measure the variation between heartbeats to determine your respiratory rate.

(https://ouraring.com/blog/respiratory-rate/?srsltid=AfmBOoobbOlfrxLvTFbkuy-tEKmYMpzNKcjg_oBbdNQTNHWAK1EUc_B)

Software

Oura Ring 4 with active membership includes:

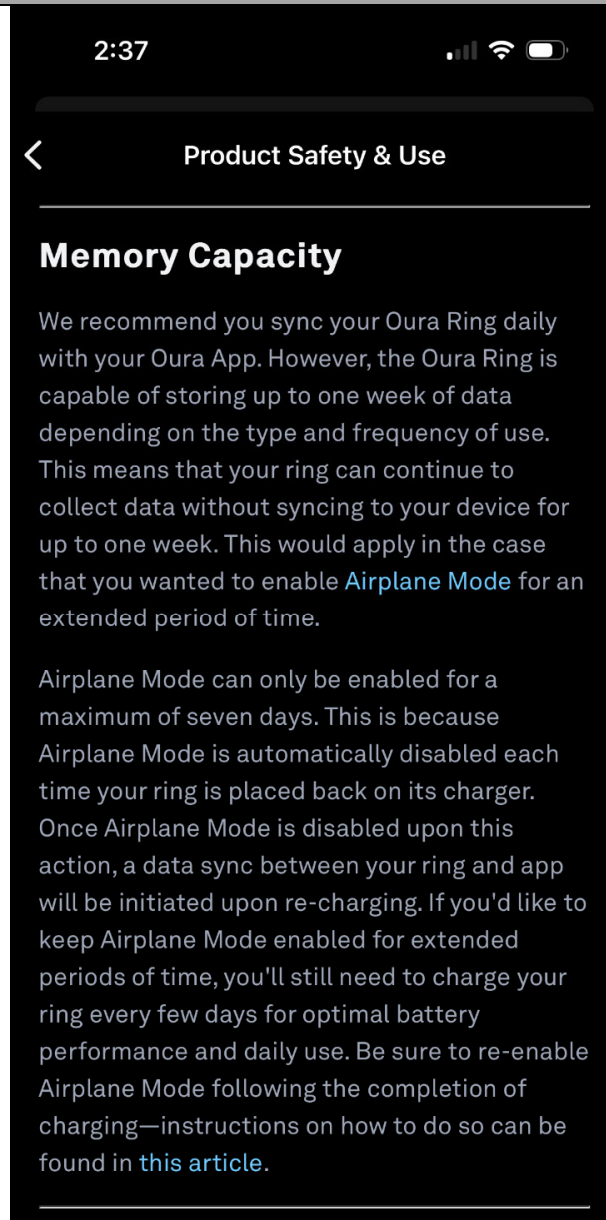
- Detailed sleep analysis
- 24/7 heart rate tracking (Daytime, Nighttime, Activity)
- Integration with third-party health and wellness apps like Strava
- Advanced temperature monitoring
- Blood Oxygen Sensing (SpO2)
- Daytime Stress
- Dynamic activity goals with Automatic Activity Detection
- Heart Health (Cardiovascular Age, Cardio Capacity (VO2 Max))
- Metabolic Health (Glucose Tracking, Meals)
- Oura Labs
- Resilience
- Weekly, monthly, quarterly, yearly, and anniversary reports
- Women's Health (Cycle Insights, Pregnancy Insights)

Body clock and chronotype, Explore content, and personalized activity goals are also available for Gen2 and Gen3 rings.

Regardless of the type of membership you have, your Oura Membership will continue to work seamlessly with Oura Ring 4. Since your membership is tied to your Oura account (not the ring itself), all of your data and benefits will carry over. Once you pair your new Oura Ring 4, you'll see all your data in the Oura App.

(<https://support.ouraring.com/hc/en-us/articles/33045011508115-Our-Ring-4>)

	U.S. Patent No. 10,642,359
14[c][v] perform a specified function using the valid signal.	The Oura Accused Product performs a specified function using the valid signal. For example, the Oura Accused Product performs a specified function, such as computing, storing, and/or transmitting sleep-related data or data collected while the user is determined to be asleep, using the valid signal.



(Oura App → Help → Product Safety & Use → Memory Capacity)

Memory Capacity

We recommend you sync your Oura Ring daily with your Oura App. However, the Oura Ring is capable of storing up to one week of data depending on the type and frequency of use. This means that your ring can continue to collect data without syncing to your device for up to one week. This would apply in the case that you wanted to enable **Airplane Mode** for an extended period of time.

Airplane Mode can only be enabled for a maximum of seven days. This is because Airplane Mode is automatically disabled each time your ring is placed back on its charger. Once Airplane Mode is disabled upon this action, a data sync between your ring and app will be initiated upon re-charging. If you'd like to keep Airplane Mode enabled for extended periods of time, you'll still need to charge your ring every few days for optimal battery performance and daily use. Be sure to re-enable Airplane Mode following the completion of charging—instructions on how to do so can be found in **this article**.

(https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

	U.S. Patent No. 10,642,359
	Oura employs data encryption, which ensures secure pairing of your ring and app. This means your personal data will be kept fully secure in case an external party ever gets a hold of your ring. The Oura App is available on the App Store and Google Play. The Oura App is supported on iOS devices running Apple iOS 15 and higher, or Android devices running Android 8.0 and higher with Google Play services. Your device must support Bluetooth 4.0. Ring firmware updates are accessible directly from your Oura App. (https://support.ouraring.com/hc/en-us/articles/360025428394-Product-Safety-Use#h_01G371BFQ5FGF4QPJ63T6AJFZZ)

How Sleep Stages Are Measured

Oura uses data from many of your biosignals — including movement, skin temperature, resting heart rate, heart rate variability (HRV), and respiratory rate — to determine which sleep stage your body is in. Here are some of the indicators that Oura looks for at each stage:

Light sleep:

- Respiratory rate slows
- Heart rate decreases
- Average body temperature drops

Deep sleep:

- Low movement
- Heart rate and respiration stabilize
- Skin temperature is at its highest

REM sleep:

- Respiratory rate increases
- Heart rate increases

Learn more about how Oura measures sleep in this video:

(<https://support.ouraring.com/hc/en-us/articles/11752397946003-Sleep-Stages>)

How Nap Detection Works

For Oura to detect sleep as a nap, it must be between 15 minutes and three hours in length, and your body must fall into at least one **sleep stage** (e.g. light, deep, or REM) during that time. It is possible that your time in bed may extend beyond 15 minutes, but changes in your heart rate, movement, and average body temperature do not yet indicate that you've physically entered sleep. Extra movement or an elevated heart rate may also delay detection of a nap. In these cases, a presumed nap may be instead labeled as a rest and shown on the **24-hour heart rate graph** as restorative time (green).

Any sleep picked up by Oura that's longer than three hours will be viewed as your longest sleep period, and added automatically.

(<https://support.ouraring.com/hc/en-us/articles/1500009653181-Nap-Detection>)

At Oura, we're committed to innovating and improving our health sensing algorithms through rigorous scientific processes, as well as sharing the research and validation with our community. Oura's Sleep Staging Algorithm 2.0, which matches performance with the gold standard [polysomnography \(PSG\) sleep lab test](#).

The result of more than two years of extensive research and development, this updated algorithm is based on a 2021 research study published in [Sensors](#) (a leading international, peer-reviewed, and open-access journal on the science and technology of sensors), which showcases the results of the algorithm.

How Accurately Does Oura Measure Sleep?

By leveraging every signal from the Oura Ring (movement, temperature, heart rate, and HRV), the *Sensors* study found Oura's sleep staging algorithm achieves **79% agreement** with gold-standard polysomnography (PSG) for classifying the [four stages of sleep](#) (wake, light, deep, and rapid eye movement (REM) sleep). By contrast, most commercial wrist-based wearables are limited to [60-65% agreement](#).

Oura's new sleep staging algorithm was developed using advanced machine learning techniques and one of the largest wearable sleep datasets collected to date. This one-of-a-kind dataset includes diverse participants across multiple continents and sleep laboratories, demonstrating the power of using all of the signals of the Oura Ring to maximize accuracy.

Read [the full paper in Sensors](#) and see key takeaways below.
(<https://ouraring.com/blog/new-sleep-staging-algorithm/>)

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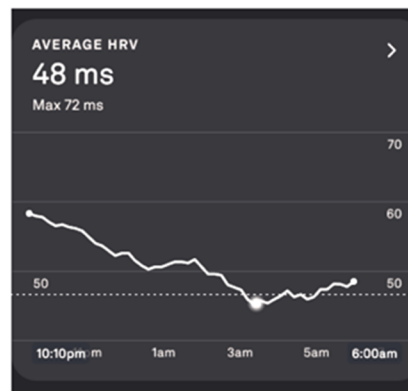
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